
Notes for ‘Composite functions’

Important Ideas and Useful Facts:

- (i) **Composition of functions:** We can build new functions out of old ones using *composition*: if f and g are functions then the *composite* function $f \circ g$ is defined by the rule

$$y = f \circ g(x) = (f \circ g)(x) = f(g(x)) ,$$

whenever this combined rule makes sense.

(Some functions are not compatible for composition, or only compatible for certain inputs and outputs, as illustrated below.)

For example, suppose that f , g and h are functions with rules

$$f(x) = 2x - 3, \quad g(x) = x^2 \quad \text{and} \quad h(x) = \frac{x+3}{2} .$$

Then

$$(f \circ g)(x) = f(g(x)) = f(x^2) = 2x^2 - 3 ,$$

whilst

$$(g \circ f)(x) = g(f(x)) = g(2x - 3) = (2x - 3)^2 = 4x^2 - 12x + 9 .$$

Notice that the rules for $f \circ g$ and $g \circ f$ are very different. However, we have

$$(f \circ h)(x) = f(h(x)) = f\left(\frac{x+3}{2}\right) = 2\left(\frac{x+3}{2}\right) - 3 = x + 3 - 3 = x ,$$

and

$$(h \circ f)(x) = h(f(x)) = h(2x - 3) = \frac{(2x - 3) + 3}{2} = \frac{2x}{2} = x .$$

Thus, in fact, $f \circ h$ and $h \circ f$ are the same function, namely, the *identity function* that inputs x and outputs x without any change. Functions with this property, that their compositions in either order produce identity functions, are said to be *inverses* of each other, a property that is studied in formal detail later.

- (ii) **Order of composition is important:** We are used to the property that we can add or multiply two numbers together in either order without making any difference. Composition of functions, however, is an operation that is very sensitive to order.

In the example above, $f \circ g$ and $g \circ f$ produced rules which were very different. This is typical: almost all of the time $f \circ g$ and $g \circ f$ will turn out to be different functions.

The example above where the composites $f \circ h$ and $h \circ f$ coincide (in fact with the identity function) is rare (though important when it happens).

Even the domain of definition of functions becomes sensitive to order under composition. For example, suppose the rules for f and g are

$$f(x) = -x^2 - 1 \quad \text{and} \quad g(x) = \sqrt{x} ,$$

where the domain of f is all of $\mathbb{R}$, but the domain of g is the interval $[0, \infty)$ of nonnegative real numbers. Then

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{x}) = -(\sqrt{x})^2 - 1 = -x - 1 ,$$

for all nonnegative real numbers x . However

$$(g \circ f)(x) = g(f(x)) = g(-x^2 - 1) = \sqrt{-x^2 - 1}$$

is not defined for any real numbers x , since $-x^2 - 1$ is always negative, and it is not possible to take the square root of a negative number (working over the real number system). The domain of $f \circ g$ is the interval $[0, \infty)$, whilst the domain of $g \circ f$ is the empty set $\emptyset$.